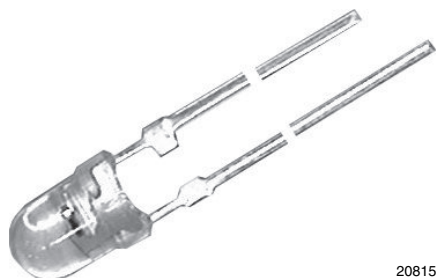


Ambient Light Sensor



FEATURES

- Package type: leaded
- Package form: T-1
- Dimensions (in mm): $\varnothing$ 3
- High photo sensitivity
- Adapted to human eye responsivity
- Angle of half sensitivity: $\varphi = \pm 30^\circ$
- Material categorization:
for definitions of compliance please see
www.vishay.com/doc?99912



RoHS
COMPLIANT
HALOGEN
FREE
GREEN
(5-2008)

DESCRIPTION

TEPT4400 ambient light sensor is a silicon NPN epitaxial planar phototransistor in a T-1 package. It is sensitive to visible light much like the human eye and has peak sensitivity at 570 nm.

APPLICATIONS

- Ambient light sensor for control of display backlight dimming in LCD displays and keypad backlighting of mobile devices and in industrial on / off-lighting operation
- Replacement of CdS photoresistors

PRODUCT SUMMARY

COMPONENT	I_{PCE} (μA)	φ (deg)	$\lambda_{0.5}$ (nm)
TEPT4400	200	± 30	440 to 800

Note

- Test condition see table "Basic Characteristics"

ORDERING INFORMATION

ORDERING CODE	PACKAGING	REMARKS	PACKAGE FORM
TEPT4400	Bulk	MOQ: 5000 pcs, 1000 pcs/bulk. Label with I_{PCE} group on each bulk. Specifications of group A / B / C see table "Type Dedicated Characteristics" on page 2	T-1

Note

- MOQ: minimum order quantity

ABSOLUTE MAXIMUM RATINGS ($T_{amb} = 25^\circ C$, unless otherwise specified)

PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Collector emitter voltage		V_{CEO}	6	V
Emitter collector voltage		V_{ECO}	1.5	V
Collector current		I_C	20	mA
Power dissipation	$T_{amb} \leq 55^\circ C$	P_V	100	mW
Junction temperature		T_j	100	$^\circ C$
Operating temperature range		T_{amb}	-40 to +85	$^\circ C$
Storage temperature range		T_{stg}	-40 to +100	$^\circ C$
Soldering temperature	$t \leq 3$ s	T_{sd}	260	$^\circ C$
Thermal resistance junction / ambient	JESTD 51	R_{thJA}	300	K/W

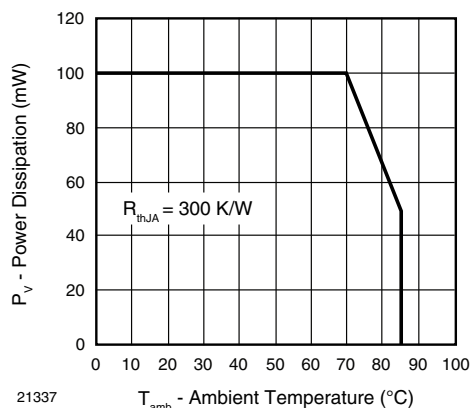


Fig. 1 - Power Dissipation Limit vs. Ambient Temperature

BASIC CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)						
PARAMETER	TEST CONDITION	SYMBOL	MIN.	TYP.	MAX.	UNIT
Collector emitter breakdown voltage	$I_C = 0.1\text{ mA}$	V_{CEO}	6	-	-	V
Collector dark current	$V_{CE} = 5\text{ V}$, $E = 0$	I_{CEO}	-	3	50	nA
Collector emitter capacitance	$V_{CE} = 0\text{ V}$, $f = 1\text{ MHz}$, $E = 0$	C_{CEO}	-	16		pF
Collector light current	$E_v = 20\text{ lx}$, CIE illuminant A, $V_{CE} = 5\text{ V}$	I_{PCE}	15	-	70	μA
	$E_v = 100\text{ lx}$, CIE illuminant A, $V_{CE} = 5\text{ V}$	I_{PCE}	-	200	-	μA
Angle of half sensitivity		ϕ	-	± 30	-	deg
Wavelength of peak sensitivity		λ_p	-	570	-	nm
Range of spectral bandwidth		$\lambda_{0.5}$	-	440 to 800	-	nm
Collector emitter saturation voltage	$E_v = 20\text{ lx}$, CIE illuminant A, $I_{PCE} = 1.2\text{ }\mu\text{A}$	V_{CEsat}	-	0.1	-	V

TYPE DEDICATED CHARACTERISTICS						
PARAMETER	TEST CONDITION	BINNED GROUP	SYMBOL	MIN.	MAX.	UNIT
Photo current	$E_v = 20\text{ lx}$, CIE illuminant A, $V_{CE} = 5\text{ V}$, $T_{amb} = 25\text{ }^{\circ}\text{C}$	A	I_{PCE}	15	28.4	μA
		B	I_{PCE}	23.5	44.6	μA
		C	I_{PCE}	36.9	70	μA

Note

- Each 5000 piece bag will contain a single group. The label on the bag will indicate which binned group is in the bag. A specific group cannot be ordered. Production shipments containing multiple bags will likely include multiple groups. Please design accordingly.

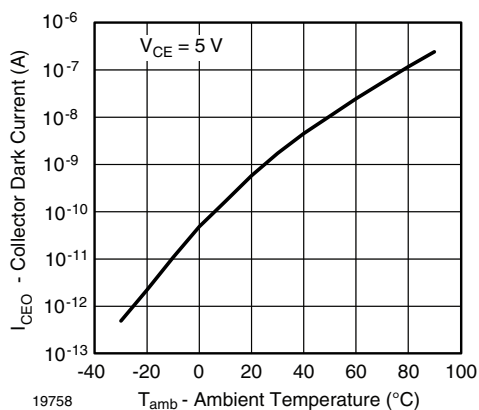
BASIC CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)


Fig. 2 - Collector Dark Current vs. Ambient Temperature

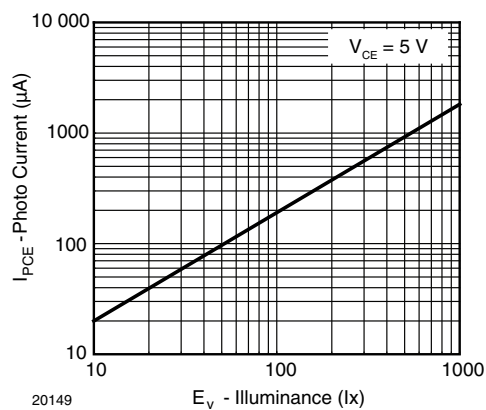


Fig. 5 - Photo Current vs. Illuminance

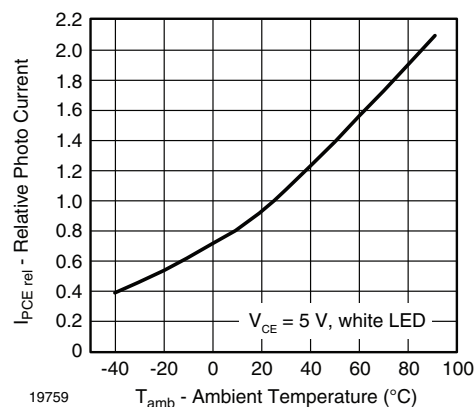


Fig. 3 - Relative Photo Current vs. Ambient Temperature

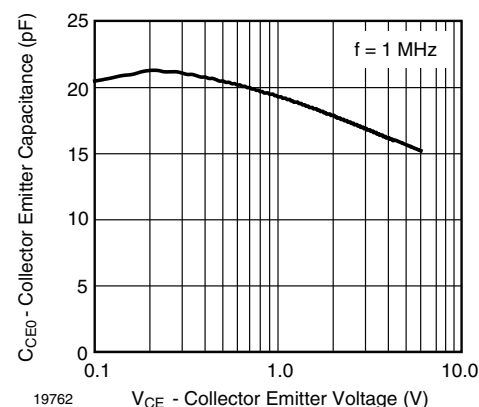


Fig. 6 - Collector Emitter Capacitance vs. Collector Emitter Voltage

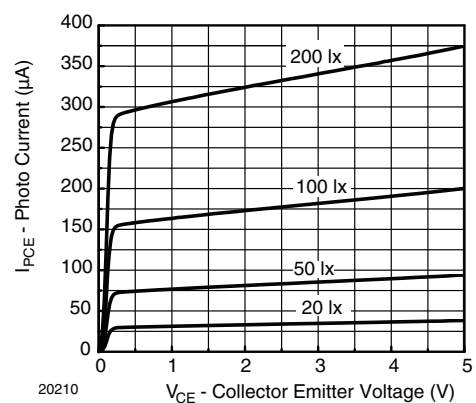


Fig. 4 - Photo Current vs. Collector Emitter Voltage

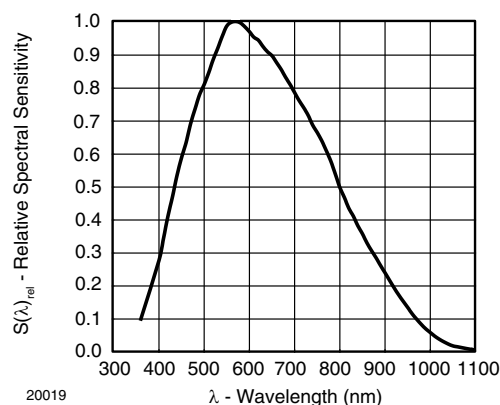


Fig. 7 - Relative Spectral Sensitivity vs. Wavelength

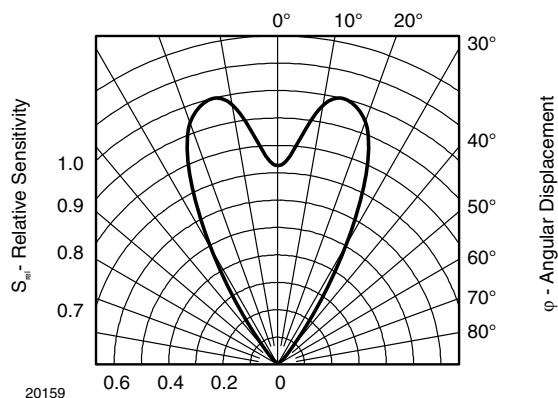
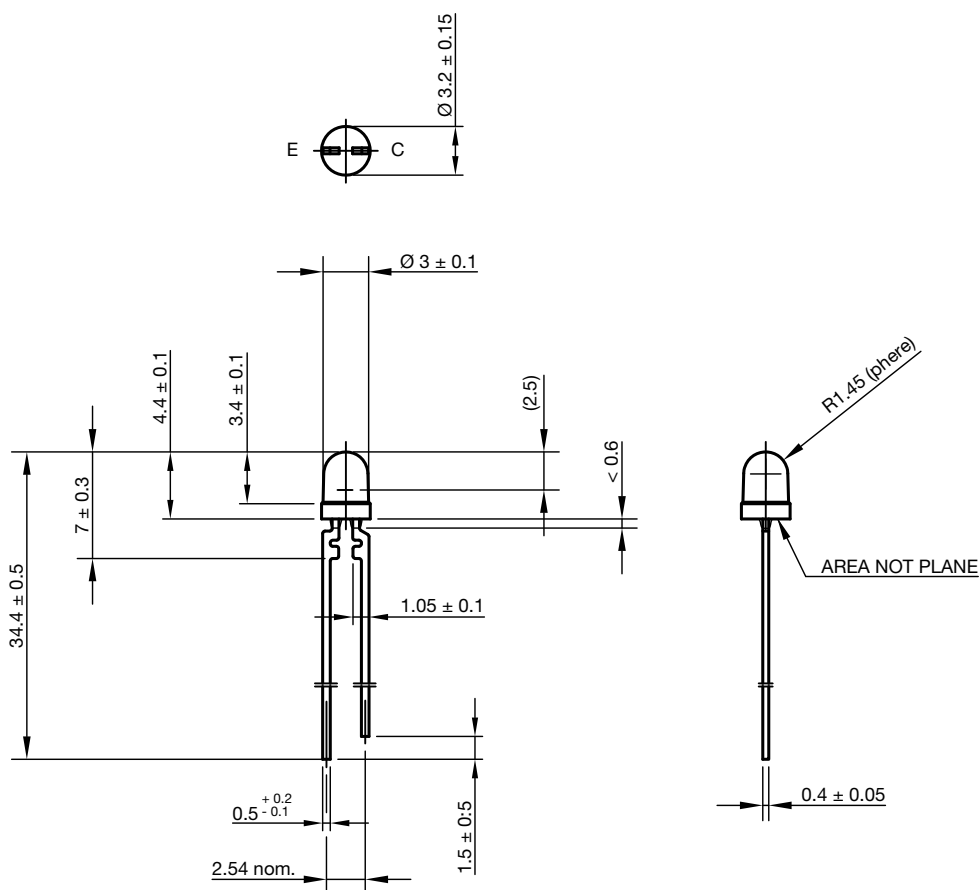
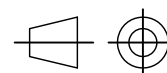


Fig. 8 - Relative Radiant Sensitivity vs. Angular Displacement

PACKAGE DIMENSIONS in millimeters



Drawing-No.: 6.544-5054.01-4
Issue: 5; 28.07.14



technical drawings
according to DIN
specifications



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